Calibrated Land Model Intercomparison Project (CalLMIP)

Phase 1 Protocol v2

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Edits in v2 since v1.2: please see all red sub-headings and text below. Calibration and validation sites selected with PFT and soil info (see Section 2.3; Note: PFT and soil information now in protocol in Section 2.3, not in flux files); changes to variable names and required information in output files (Section 6) and associated changes to file name (Section 7); changes to protocol for uploading files to ME-org (mostly Section 11.5); other minor changes (see red text); added checklist to help planning; added TOC to help navigation; revised timeline.

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Background

CalLMIP is a new community-led initiative to evaluate how well models' existing ("in-house") parameter calibration efforts reduce model-data misfit and the spread across models for key variables. CalLMIP is led by members of the Analysis and Integration of Modeling Earth Systems (AIMES) Land Data Assimilation Working Group and the International Land Modeling Forum (ILMF) Parameter Estimation Working Group. Please see the CalLMIP website (https://callmip-org.github.io) "About" page for more information on the CalLMIP steering committee.

This initiative is targeted at global-scale land models – i.e., prognostic, process-based land surface or terrestrial biosphere models that form part of Earth system models, or intermediate complexity models and vegetation demographic models that are routinely run at global scale (for example for the global carbon budget).

The overall goals envisaged for CalLMIP are:

- To quantify the reduction in individual model uncertainty and inter-model spread on key optimised and non-optimised variables (e.g., carbon and water stores and fluxes, aboveground biomass, soil carbon) as a result of parameter calibration.
- To encourage the use of uncertainty quantification methods for constraining parameter uncertainty in global land models.
- To better understand and exchange knowledge on the variety of methods being

used in global land model parameter calibration in order to identify best practices.

In Phase 1 we are taking a *'come as you are'* approach, which is to leverage existing/current model parameter estimation systems, rather than requiring development of new systems or tests of different calibration configurations. With this approach, we hope to maintain flexibility to include a wide array of models with their existing calibration systems.

In Phase 1 we will also have a focus on **site-level calibration** focusing only on the most common observations measured at flux tower sites: net ecosystem exchange (**NEE**), latent heat flux (**LE**) and sensible heat flux (**H**). Other key ecosystem variables – such as LAI, above and belowground carbon stocks and soil moisture – will be the target of future phases of CalLMIP. Regional to global scale (as well as coupled) parameter calibration experiments will also be a target for future CalLMIP Phases.

Our hope is that Phase 1 is a starting point from which additional analyses and experiments will develop. We will outline the vision for future CalLMIP phases from the CalLMIP Steering Committee and discussion at the 1st Phase Planning workshops held in March and April 2025 in a perspective article to be submitted in spring 2026.

Key Information

- CalLMIP website: https://callmip-org.github.io
    - Brief overview of everything CalLMIP, including links to Phase 1 protocol, data, and event information.
- CalLMIP GitHub site: https://github.com/callmip-org/Phase1
    - Phase 1 repository containing protocols and data (so far). Analyses will be shared via this repository as well as on the CalLMIP workspace on modelevaluation.org.
- CalLMIP workspace on modelevaluation.org
    - Information on experiments, sites included in each experiment, forcing data and participating models. Model outputs will be uploaded here.
- AIMES Land DA Working Group website: https://aimesproject.org/cal-lmip/
    - For recordings from the 1st Phase Planning and Development Workshop held in March and April 2025.

Phase 1b Checklist

- ☐ 1. Create a model profile on ME-org if not already done (see Section 11.2)
- ☐ 2. Download met forcing files from ME-org for both Phase 1b Calibration and Phase 1b Validation experiments (see Section 2.3 for sites and Sections 11.3 and 11.4 for instructions on ME-org)
- ☐ 3. Download flux observation files for the multi-site calibrations from the CalLMIP GitHub Phase 1b data repository here: https://github.com/callmip-org/Phase1/tree/main/Data/Phase1b
- ☐ 4. Use atmospheric CO_2 concentration from the TRENDY file on the GitHub repository here starting in year 1850 for spin-up: https://github.com/callmip-org/Phase1/tree/main/Data/Non-site-specific_forcing
- ☐ 5. If not using prognostic LAI, use LAI from the met file downloaded from ME-org (see Section 4 for further details)
- ☐ 6. Individual model groups decide whether to include N cycling and LUC and, if so, which datasets to use, as LUC and N deposition and fertilisation are not provided for site level runs.
- ☐ 7. Download example "template" output .nc file from the main CalLMIP Phase 1 GitHub repository here: https://github.com/callmip-org/Phase1/tree/main/template_output_file). This file is named according to the convention specified in Section 7 and also contains all variables, names, units and global attributes specified in Section 7 of the protocol (and see checklist in Step 8 below).
- ☐ 8. Run the model with prior values for the 21 calibration sites (see Section 2.3)
    - ☐ Check correct naming format for nc output file: ….nc (see Section 7 and see example template .nc file)
    - ☐ Check output files contain all information (check Section 6) and that the output file matches the example template .nc file
    - ☐ Check file contains full list of variables (see Section 6.2)
    - ☐ Check using correct variable names and units as in Table 3 in Protocol (Section 7) and the template .nc file
    - ☐ Check providing uncertainties (or ensemble simulations) on variables optimised (e.g., NEE, LE and H for Calibration Scenario 1)
    - ☐ Check Time variable is correct (see Section 6.2 and verify using template .nc file):
        - ☐ units: "*days since YYYY-01-01 00:00:00*", YYYY is the start year from the met file
        - ☐ Time steps are 1, 2, 3,... or 0.5, 1.5, 2.5,...
    - ☐ Check the following for lat and lon (see Section 6.2 and verify using template .nc file)
        - ☐ Added as variables
        - ☐ Are scalar (individual numerical) values (e.g., floats, not strings or arrays) when printed

- [ ] Variables are 3D <*time, lat, lon*>

- [ ] Add the following to global attributes (see Section 6.3 and verify using template .nc file):
  - [ ] Model: <model_name>.<model_version>
  - [ ] CalLMIP_Phase: Phase1b
  - [ ] Calibration_Scenario: <Scen1><Scen2><Scen3><Scen4><Scen5><Scen6>
  - [ ] Calibration_stage: <Prior> or <Posterior>
  - [ ] Cal_Val: <Calibration> or <Validation>
- [ ] Create a new model output in the Phase 1b Calibration experiment on ME-org for your prior simulations for all sites (i.e., 21 .nc files)
  - [ ] Follow steps in Section 11.5 for steps on how to upload a set of model simulations and how to run the benchmarking against empirical models

- [ ] 9. Perform multi-site calibration across the 21 sites **for Calibration Scenario 1** (NEE, LE and H included in the calibration).
  - [ ] Check output files match the example template .nc file and follow same checklist as in Step 8.
  - [ ] Create a new model output in the Phase 1b Calibration experiment on ME-org for your Scenario 1 posterior simulations for all sites (i.e., 21 .nc files)
    - [ ] Follow instructions in Section 11.5 for how to upload a set of model simulations and how to run the benchmarking against empirical models

- [ ] 10. Perform prior runs for 10 validation sites. Repeat Step 8 but with following differences:
  - [ ] Have “Val” in filename instead of “Cal”
  - [ ] Make sure the “Cal_Val” global attribute keyword is set to “Validation” in the output file
  - [ ] When you upload model outputs:
    - [ ] have “Val” in the “Name” of the Model output; and
    - [ ] Add the runs to the Phase 1b Validation experiment on ME-org, not Phase 1b Calibration

- [ ] 11. Run validation runs for 10 validation sites using calibrated parameters. Repeat Step 9 above with following differences (as in Step 10):
  - [ ] Have “Val” in filename instead of “Cal”
  - [ ] Make sure the “Cal_Val” global attribute keyword is set to “Validation” in the output file
  - [ ] When you upload model outputs:
    - [ ] have “Val” in the “Name” of the Model output; and
    - [ ] Add the runs to the Phase 1b Validation experiment on ME-org, not Phase 1b Calibration

- [ ] OPTIONAL: Repeat Steps 9 and 11 above for Calibration Scenarios 2-6. There is no need to repeat the prior runs. Be sure to change the filename, global attribute “calibration_Scenario” keyword and “Name” in the Model Output upload to the correct scenario number.

- [ ] Total number of .nc files for *calibration*: 21 Prior (uploaded as one model output) + 21 posterior for Calibration Scenario 1 (NEE/LE/H) (uploaded as a second model output) + … [21 posterior for each additional optional calibration scenarios (posteriors from each scenario uploaded as a separate model output)]
- [ ] Total number of .nc files for *validation*: 10 Prior (uploaded as one model output) + 10 posterior for Calibration Scenario 1 (NEE/LE/H) (uploaded as a second model output) + … [10 posterior for each additional optional calibration scenarios (posteriors from each scenario uploaded as a separate model output)]
- [ ] Total number of model outputs in ME-org: 1 Cal/Prior, 1 Val/Prior, 1-6 Cal/Posterior, 1-6 Val/Posterior, i.e., a minimum of 4 and a maximum of 14 Model outputs per model, linked to relevant Calibration Scenarios.

Timeline for CalLMIP Phase 1 (revised in v2)

- March/April 2025: Three community sessions to decide on protocol, data, sites etc.
- May-August 2025: Preparation of input and observation files and data harmonization by the MacBean lab for all modeling groups.
- November 2025: Deliver inputs and finalised protocol to modeling groups.
- November 2025: Webinar to the launch CalLMIP Phase 1.
- **November 2025 to early January 2026: Read through Phase 1a protocol and decide whether you would like to participate. If so, please fill out the Participant Form by January 9th 2026** (available on the CalLMIP website (https://callmip-org.github.io) under the 'Protocols and Data' → Phase 1 → Participants.
    - Draft of Phase 1 protocol paper circulated to all participating model groups for details on model and calibration set-up.
- **November to March 2026: Phase 1a "test" calibration at 1 site – model prior and posterior simulations post calibration uploaded to modelevaluation.org by February 27th 2026.**
- **June to mid-August 2026: Full multi-site Phase 1 parameter calibration experiments/scenarios carried out by individual modeling groups – model prior and posterior simulations uploaded to modelevaluation.org by mid-August 2026 if you'd like your group's results to be presented at LSMS2 (see below)**
- Mid-August: submission of Phase 1 protocol paper to GMD.

- September 2026: Presentation of initial site level calibration results at the 2nd Land Surface Modeling Summit (LSMS2).
- Fall 2026:
    - Global scale simulations with calibrated parameters. Calibrated model simulations uploaded to modelevaluation.org by October 30th 2026.
    - Initial analysis and draft paper for site-level calibration intercomparison in discussion/collaboration with all participating model groups.
- December 2026: Presentation of site-based calibrations and initial results from global post-optimization simulations at AGU 2026.
- January to April 2027:
    - Finalise site level intercomparison paper and submit.
    - Draft paper for global post-optimization intercomparison
    - Planning for possible Phase 2.

Protocol Plan

1. Overview of CalLMIP Phase 1

In this section we outline some of the components of the calibration experiments that will be "fixed" or constant across all participating model groups followed by components that are "flexible" and can be decided by each group. We note that in an ideal world many more components would be constant across model groups; however, as noted in the Background section, in Phase 1 we are taking a *'come as you are'* approach that we hope will enable as many model groups as possible to participate. In future phases of CalLMIP we hope to test differences between calibration systems, forcing data, parameters selected, data types included, etc. We will outline the vision for future CalLMIP phases from the CalLMIP Steering Committee and discussion at the 1st Phase Planning workshops held in March and April 2025 in a perspective article to be submitted in spring 2026.

1.1 Elements that will be constant across all experiments and modeling groups

To ensure comparability across modeling groups, certain key aspects of the calibration experiment will be standardized, including:

- No. of sites (see Section 2).
- Required calibration experiments/scenarios – including an initial "Phase1a" test calibration at 1 site (see Section 3).
- Flux tower data: forcing datasets and flux observations (variable type and their associated measurement errors) – including common record length across all groups for calibration and validation (see Sections 4 and 5).
- The required output variables will be standardized across all models (see Section 6).
- Posterior uncertainties on parameters and variables optimized will be required from all models (see Section 6).
- Output file format will be standardized across models (see Section 7).
- Spin-up protocol will be partly standardized across all groups (see Section 8) for further details).
- Calibration/validation procedure (both spatial and temporal validation) will be standardized (see Section 9).

- Post optimization global simulations will be required by all groups after the full multi-site calibration (see Section 10).
- All methods and choices must be thoroughly documented by all participating groups in the Phase 1 protocol draft article (see Section 12 for information required).

Further information on all required elements is provided in the following Sections.

1.2 Elements that will be defined by each modeling group

While the core structure of the CalLMIP experiments will be standardized (see Section 1.1), modeling groups will have flexibility in defining certain methodological aspects. These “flexible” elements of the calibration experiments are detailed in the bullet points below in this section and not discussed further in the protocol:

- Model version: each group will select which version of their model they want to submit to CalLMIP. This includes choices such as whether LAI is prescribed or treated as a prognostic variable, whether N cycling is included, whether LUC is included, and how to translate PFT type/fractional cover and soil texture information from each site to fit their model. If a group runs a variant of a land surface model that contributed to CMIP6, and can run a CMIP6-like configuration, and/or if they have the version for CMIP7-FT, we ask that one or both of these versions be prioritized.
- Parameter selection: each group will decide which and how many parameters they want to optimize and which process parameters they want to include in the calibration. Parameters may vary spatially (e.g., by PFT) but should be time-invariant. Participating modeling groups can also choose their method for parameter selection (e.g., expert elicitation, sensitivity analysis). Furthermore, each group will determine what prior parameter bounds and uncertainties to use for their parameter calibration experiments.
- Parameter calibration method: each group will select its own method for calibration (e.g., gradient-based optimization, MCMC approaches, or machine learning techniques).
- Cross/multi-site parameter approach: parameter optimization must result in a set of “operational” parameters (e.g., for running the model globally), rather than site-specific parameter sets, although PFT-specific and soil-specific parameters, etc, are allowed. The specific method for this implementation is left to each group’s discretion (e.g., hierarchical Bayesian models, individual site calibrations followed by averaging, multi-site optimizations for each PFT, parameter ensembles etc).
- The method for including multiple data types in the calibration experiment (e.g., one at a time/stepwise or all-together/simultaneous) is left to the group’s

discretion.

- Number of PFTs to be calibrated for each site: each group will decide whether they calibrate the parameters for all PFTs at a given site or only for the most dominant PFT, depending on model configuration and parameter calibration system.
- Model errors are specific to each model and need to be estimated by each group and combined with measurement errors provided, as appropriate for the parameter calibration method.
- Optional calibration scenarios: groups can opt into running the additional, optional Calibration Scenarios 2 to 6 (see Section 3.1).
- Certain elements of the spin-up are left to each participating model group to decide (see Section 8).

NOTE: the decisions modeling groups make for each of these points must be documented in the Phase 1 protocol draft article, which will be sent to all modeling groups that sign-up to participate (see Section 12 for further information).

2. Site selection

2.1 Overview

We have selected 21 sites from the 42 sites originally selected from the PLUMBER2 list of sites. Below is a description of the site selection process and the sites selected.

For each site, the following are provided: i) gap-filled *in situ* meteorological forcing data (variables typically used in land surface models – see Section 4) from the flux tower for the entire calibration and validation period; ii) measured (i.e., “raw” or “uncorrected”) daily observations of NEE, LE, and H, with measurement errors, for calibration only (see Section 5); iii) PFT type and % cover; iv) soil layer depth and texture information; v) atmospheric CO_2 forcing data for the observation period.

2.2 Site Selection Procedure

The sites used in the first phase of CalLMIP will be based on the PLUMBER2 dataset (Abramowitz et al., 2024). PLUMBER2 provides pre-selected sites from FLUXNET2015 (CC-BY-4.0 licensed sites), FLUXNET La Thuile Free-Fair-Use subset and OzFlux data, that meet quality control criteria and have minimal data-sharing restrictions. PLUMBER2 quality control steps ensure that all sites have complete metadata, including canopy height and IGBP vegetation type, and contain full years of data with minimal missing values for key variables. Furthermore, standardized gap-filling methods (e.g., LWdown from Abramowitz et al. (2012), PSurf from elevation and temperature) and FLUXNET2015-aligned energy balance corrections have been applied to these sites as part of the PLUMBER2 processing steps; however, we will use only measured energy fluxes for model calibration within CalLMIP (see Section 5).

From the original 170 PLUMBER2 sites, an initial 42 sites from PLUMBER2 were chosen by Gab Abramowitz and colleagues for their own model development pipeline. The main criteria for selecting these sites were: (a) length of record (meaning model spin-up issues were less likely to interfere with evaluation), and (b) data quality (less gap-filling of key driving variables - precipitation, temp, radiation, humidity, wind) (G. Abramowitz, personal communication, March 26th, 2025).

We added the following additional criteria for site selection within CalLMIP based on feedback from the CalLMIP 1st Phase planning workshops held in March and April 2025 and decisions made by the CalLMIP steering committee:

1. Four year minimum record length. Within these four years there had to be at least three years worth of daily data after taking into account missing and gap-filled data with no large gaps in coverage. Only gap-filled data from NEE, LE, and H was assessed to determine if there was an appropriate amount of data.
2. Netcdf files had to include uncertainties for NEE, LE, and H. In the PLUMBER2 archive, only netcdf files using FLUXNET 2015 contained uncertainties. Australian sites processed using OzFlux provided no uncertainties, therefore these sites were excluded for now. Some older sites were from the LaThuille dataset, which only had uncertainties for NEE and were therefore excluded.
3. Some sites were further excluded because Qg (ground heat flux) was not measured. Though Qg is not a mandatory measurement within the Fluxnet processing pipelines Qg is essential for calculations of energy balance closure. EBC allowed for the calculation of an energy balance correction factor that was then used in an equation to calculate the joint uncertainty.
4. PFT and soil texture information had to be available from at least one source (see Section 2.3).
5. No crop or wetland sites were included in Phase 1.

2.3 Selected sites

2.3.1 Calibration sites

Of the 42 pre-selected PLUMBER2 sites, the 21 sites listed in Table 1 (and shown in Figure 1) met our selection criteria for the *calibration* sites. (Note: DK-Sor was the test site for Phase 1a but should still be included in the multi-site calibration in Phase 1b). Please see the appendices of the Phase 1 protocol paper for further information on sites and sources for site information.

Site	Record length used (years)	PFT type and % cover	Soil Layer Depth (cm)	Percentage of Sand, Silt, Clay
CA-Qfo	7	100% needleleaf evergreen tree (boreal)	1. 27.5	1. 58, 32, 10
CH-Dav	18	100% needleaf evergreen tree (boreal)	1. 0-24	1. 58, 32, 10
DE-Gri	11	100% C3 grass (temperate)	1. 0-23 2. 23-70	1. 10, 81, 9 2. 13, 76, 11
DE-Hai	13	96.5% broadleaf deciduous tree (temperate); 3.5% C3 grass (temperate)	1. 0-50	1. 4, 56, 40
DE-Tha	17	87% needleaf evergreen tree (temperate); 10% needleleaf deciduous tree (temperate); 3% broadleaf deciduous tree (temperate)	1. 0-120	1. 17, 71, 12
DK-Sor	18	80% broadleaf deciduous tree (temperate); 20% needleleaf evergreen tree (temperate)	1. 0-4.5 2. 4.5-9.1 3. 9.1-16.6 4. 16.6-28.9	1. 60, 27, 13 2. 63, 26, 12 3. 63, 25, 12 4. 62, 24, 13
FI-Hyy	19	100% needleleaf evergreen tree (boreal)	1. 0-5 2. 5-26 3. 26-60	1. 52, 39.5, 8.5 2. 51, 40, 9 3. 48, 41, 11
FR-Pue	15	90% broadleaf evergreen tree (temperate); 10% broadleaf deciduous tree (temperate)	1. 0-50	1. 14.1, 46.3, 39.6
IT-Lav	10	85% needleleaf evergreen tree (temperate) 15% broadleaf deciduous tree (temperate)	1. 0-4.5 2. 4.5-9.1 3. 9.1-16.6 4. 16.6-28.9	1. 46, 38, 16 2. 45, 39, 16 3. 58, 29, 13 4. 58, 30, 11
IT-Mbo	10	100% C3 grass (temperate)	1. 0-4.5 2. 4.5-9.1 3. 9.1-16.6 4. 16.6-28.9	1. 46, 38, 16 2. 45, 39, 16 3. 58, 29, 13 4. 58, 30, 11

IT-Noe	11	80% broadleaf evergreen shrub (temperate); 13% biocrust; 7% bare soil *(for models without biocrust – suggest having 20% bare soil not any other vegetation)*	1. 0-35	1. 17.5, 25, 57.5
NL-Loo	17	93.6% needleleaf evergreen tree (temperate) 3.5% C3 grasses 2.9% broadleaf deciduous tree (temperate)	1. 0-20	1. 96, 1, 2
RU-Fyo	12	86% needleleaf evergreen tree (boreal) 14% broadleaf deciduous tree (boreal)	1. 0-100	1. 22, 20, 58
US-MMS	16	100% broadleaf deciduous tree (temperate)	1. 0-100	1. 34, 3, 63
US-NR1	16	83% needleleaf evergreen tree (boreal); 17% broadleaf deciduous shrub (boreal)	1. 0-15	1. 82, 12, 6
US-SRG	6	44% C4 grass; 11% broadleaf deciduous tree (temperate); 45% bare soil	1. 0-100	1. 82.0, 12.0, 6.0
US-SRM	11	35% broadleaf deciduous tree (temperate); 15% C4 grass; 50% bare soil	1. 0-20	1. 82.0, 12.0, 6.0
US-Ton	14	60% C3 grass 40% broadleaf deciduous tree (temperate)	1. 0-100	1. 42.75, 43.5, 13.75
US-Var	14	100% C3 grass (temperate)	1. 0-10 2. 10-35 3. 35-50 4. 50-75	1. 29, 58, 13 2. 29, 58, 13 3. 29, 58, 13 4. 29, 58, 13
US-Whs	7	40% broadleaf evergreen shrub (temperate); 11% broadleaf drought-deciduous shrub	1. 0-100	1. 58, 32, 10

		(temperate); 49% bare soil		
US-Wkg	10	37% C4 grass; 3% broadleaf evergreen shrub (temperate) 60% bare soil	1. 0-5 2. 0-147	1. 67, 16, 17 2. 70, 20, 10

Table 1: Record length, PFT type and % cover, soil depth and texture information for the 21 selected sites for the multi-site calibration in Phase 1b from the original 42 PLUMBER2 sites selected by Gab Abramowitz. Please see Phase 1 protocol paper for information sources.

<u>Summary of key PFTs for calibration sites (note some are counted twice if the site has mixed PFTs with PFT fraction >25%)</u>:

- 5 Boreal needleleaf evergreen forests
- 2 Temperate needleleaf evergreen forests
- 4 Temperate broadleaf evergreen forests
- 5 Temperate broadleaf deciduous forests
- 4 C3 grass
- 3 C4 grass

This includes 5 mixed PFT sites (e.g., savannas). We did not include any crop or wetland sites in Phase 1.

Unfortunately, no tropical broadleaf evergreen sites met our selection criteria. We plan on performing an updated calibration in a “Phase 2” experiment with additional sites and longer record lengths once FLUXNET Shuttle site data have been fully processed for use in LSMs (and benchmarking models are available on ME-org).

Figure 1: Location of the 21 selected sites for the multi-site calibration in Phase 1b from the original 42 selected by Gab Abramowitz

Note 1: If model groups are doing individual site simulations and would like a priority list before the initial mid-August 2026 deadline, please perform at least one calibration per PFT to start with.

2.3.2 Validation sites

Out of the remaining 21 sites not selected for calibration from the original 42 pre-selected PLUMBER2 sites, the 10 sites listed in Table 2 met our selection criteria for the validation sites.

Site	**Record length used (years)**	**PFT type and % cover**	**Soil Layer Depth (cm)**	**Percentage of Sand, Silt, Clay**
AU-ASM	7	60% broadleaf evergreen (temperate); 40% C4 grass (estimated fCover)	NA	74, 11, 15
AU-How	15	50% broadleaf evergreen tree (tropical); 25% broadleaf deciduous shrub (temperate); 25% C4 grass	1. 0-15 2. 15-23	1. 58, 32, 10 2. 58,15,27
AU-Stp	8	100% C4 grass (tropical)	1. 0-20 2. 20-89	1. 40, 40, 20 2. 30, 35, 35
AU-Tum	16	100% broadleaf evergreen tree (temperate)	1. 0-0.1m 2. 0.1-0.2m 3. 0.2-0.3m 4. 0.3-0.4m 5. 0.4-0.5m 6. 0.5-0.6m 7. 0.6-0.7m 8. 0.7-0.8m 9. 0.8-0.9m 10. 0.9-1m 11. 1-1.2m 12. 1.2-1.5m 13. 1.5-1.9m 14. 1.9-2.4m 15. 2.4-3.4m	1. 64, 17, 19 2. 61.6, 17, 21.4 3. 59.2, 17, 23.8 4. 57.693, 17.147, 25.16 5. 57.08, 17.44, 25.48 6. 56.4667, 17.7333, 25.8 7. 55.853, 18.027, 26.12 8. 55.24, 18.32, 26.44 9. 54.6267, 18.6133, 26.76 10. 54.013, 18.907, 27.08 11. 52.7867, 19.4933, 27.72

				12. 50.9467, 20.3733, 28.68 13. 48.493, 21.547, 29.96 14. 45.4267, 23.0133, 31.56 15. 39.293, 25.947, 34.76
CH-Cha	18	100% C3 grass (temperate)	1. 0-20 2. 20-30 3. 30-45 4. 45-60	1. 35.8, 45.2, 19 2. 27.8, 47.4, 24.8 3. 37.4, 37.4, 25.2 4. 68.8, 21.3, 9.9
US-FPe	7	100% C3 grass (temperate)	1. 0-100	1. 58, 32, 10
US-GLE	6	100% needleleaf evergreen tree (boreal)	1. 0-9.1 2. 9.1-16.6 3. 16.6-28.9	1. 64, 23, 13 2. 65, 22, 13 3. 65, 22, 13
US-Ha1	15	95% broadleaf deciduous tree (temperate); 5% needleleaf evergreen tree (temperate)	1. 0-0.1m 2. 0.1-0.2m 3. 0.2-0.3m 4. 0.3-0.6m 5. 0.6-1m 6. 1-26.4m	1. 42.84, 53.01, 4.15 2. 53.72, 40.96, 5.32 3. 57.35, 36.95, 5.7 4. 62.28, 32.39, 5.33 5. 66.36, 28.59, 5.05 6. 67.81, 26.97, 5.22
US-Me2	13	100% needleleaf evergreen tree (temperate)	1. 0-20 2. 20-50 3. 50-100	1. 69, 24, 7 2. 66, 27, 7 3. 54, 35, 11
US-UMB	15	95% broadleaf deciduous tree (temperate); 5% needleleaf evergreen tree (temperate)	1. 0-0.4m 2. 0.4-1m 3. 1-1.5m 4. 1.5-1.9m 5. 1.9-11.4m 6. 11.4-56.4m	1. 92, 7, 1 2. 92.5, 6, 1.5 3. 92.5, 5, 2.5 4. 92.5, 4, 3.5 5. 92.25, 4, 3.75 6. 92, 4, 4

Table 2: Record length, PFT type and % cover, soil depth and texture information for the 10 validation sites in Phase 1b selected from the 21 remaining sites not selected for calibration from the original 42 PLUMBER2 sites selected by Gab Abramowitz. Please see Phase 1 protocol paper for information sources. Note information was less readily available for these sites, particularly for the soil information (one reason why they were not included in the calibration site list). Note also that we have included one tropical site (AU-How) even though no tropical sites were included in the calibration sites because some models do not separate between biomes. Note also that the AU-ASM fCover was estimated based on available information so as to include an additional mixed site with TeBE and C4.

Summary of key PFTs for validation sites (note some are counted twice if the site has mixed PFTs with PFT fraction >25%):

- 1 Boreal needleleaf evergreen forests
- 1 Temperate needleleaf evergreen forests
- 2 Temperate broadleaf evergreen forests
- 3 Temperate broadleaf deciduous forests
- 2 C3 grass
- 3 C4 grass

This includes 2 mixed PFT sites (e.g., savannas).

Please see the appendices of the Phase 1 protocol paper for the reasons why the validation sites were not selected for calibration, as well as a list of sites that were excluded from CalLMIP Phase 1 from the entire PLUMBER2 archive and the reason why they were not included.

2.4 PFT and soil texture information

PFT and soil information for the calibration sites is provided in Table 1 and in Table 2 for the validation sites.

PFT information was not always available as % fractional cover for the typical land surface model PFTs within the PLUMBER2 netcdf files; therefore, we consulted external sources where needed. External sources included the Shi et al., (2024) dataset (Shi et al., 2025), information from flux tower site websites and publications, information from the CLASSIC CBC dataset (Meyer et al., 2026) and collaboration with Vincent Tartaglione and Cédric Bacour at LSCE who have performed their own analysis of the PLUMBER2 PFT information (more details will be provided in the Phase 1 protocol draft paper). Wherever possible we confirmed information directly from site related publications. We documented inconsistencies between different sources and the reason for those but will not provide that information here. Please see the appendices of the Phase 1 protocol paper for further details.

Information on soil depth and texture (% sand, silt and clay content) was collected from PLUMBER2 netcdf files where available, but otherwise sourced from Shi et al. (2024) or the CLASSIC CBC files (Meyer et al., 2026).

3. Calibration Scenarios

3.1 Variables included

Daily NEE, LE and H (Qle and Qh, respectively) in the flux observation file) data will be provided for all sites with their associated uncertainties. The following is a list of potential calibration scenarios with these data; however, **only the 1st calibration scenario is required** for model groups participating in CalLMIP Phase 1.

1. **NEE, LE and H (required)**
2. NEE only (optional)
3. LE only (optional)
4. H only (optional)
5. NEE and LE (optional
6. LE and H (optional)

Note 1: "CalLMIP Experiment" no. has been replaced with "Calibration Scenario" no. to avoid confusion with the CalLMIP Experiments on ME-org (Phase 1a - test, Phase 1b Calibration and Phase 1b Validation). See also Sections 6.3 and Section 7 for associated addition to global attributes and change to filename, respectively.

Note 2: while all participating groups must run Calibration Scenario 1, the method for combining all 3 datastreams in the same calibration (e.g., one at a time/stepwise or all-together/simultaneous) is left to each group's discretion.

3.2 Test calibration at 1 site (Phase 1a)

Phase 1a is designed to help each model group get the experimental set-up established for the Phase 1 experiments. For this we ask you to perform the calibration following the full protocol for just one site (DK-Sor) and upload your simulations to ME-org (see Section 11). This will also allow us to verify that ME-org processing script can read and process your model outputs.

3.3 Full multi-site calibration and spatial validation (Phase 1b)

Once all model groups have tested their CalLMIP set-up in Phase 1a, the full multi-site calibration experiments will begin in Phase 1b. See Section 2.3 for sites selected for calibration and validation

4. Forcing Datasets

A common climate forcing dataset will be used by all participating model groups to ensure consistency across sites. The forcing data files were prepared for PLUMBER2 by Gab Abramovitz.These data are the in situ flux tower meteorological measurements collected at each of the selected sites. The forcing datasets include LWdown, SWdown, Tair, Qair, Rainf, Wind and PSurf for each site, among other variables.

The met forcing files are distributed to participating modeling groups via modelevaluation.org (see Section 11). The same meteorological forcing files in NetCDF format will be provided to all groups for all selected sites. Individual modeling groups will then need to modify the files to fit their model (for example, changing the variable names, adjusting or renaming the units and other attribute information, separating snow and rain, etc).

The NCO code needed to convert Precip into Rainf and Snowf is provided in the Experiment pages for all CalLMIP phases on ME-org for those modelers who can use/need both variables. See Section 11 for how to navigate to the Phase 1a - test, Phase 1b Calibration and Phase 1b Validation experiments on ME-org.

For models that predict a prognostic leaf area index (LAI) we would prefer you to use that model capability. For models without prognostic LAI, models should use the remotely sensed LAI time series provided for each site in the PLUMBER2 netcdf forcing files. The LAI time series were derived from MODIS (8-daily MCD15A2H product, 2002-2019) or Copernicus Global Land Service (monthly, 1999-2017), though note that the data are provided for the full time period of met forcing at the site, with one of these chosen for each site, at the same time step size as meteorological forcing, based on a site-by-site analysis considering plausibility and some in-situ data (Abramowitz et al., 2024; PLUMBER2 protocol). The preferred product has a variable name “LAI”, the other “LAI alternative”. Time-varying LAI is provided for the time period covered by the remotely sensed products and otherwise a climatology constructed from all available years is used (Abramowitz et al., 2025; PLUMBER2 protocol).

Atmospheric CO_2 forcing data for the historical/transient run will also be provided via the GitHub CalLMIP site in Phase1a repository Data folder https://github.com/callmip-org/Phase1/tree/main/Data).

LUC and N deposition and fertilisation data are not provided for site level calibrations. Individual model groups decide whether to include N cycling and LUC and which datasets to include.

Any additional datasets required to run the model should be sourced by individual modelling groups and documented in the individual model sections (3.x) of the protocol paper.

5. Flux Data for Optimization

Optimization data include standard *in situ* flux tower data for optimization: **NEE, LE (Qle in flux observation file) and H (Qh in flux observation file)**. Datasets are provided at daily frequency for the selected sites for the calibration period only. One year of data has been retained for temporal validation; therefore, the flux data file is one year shorter than the meteorological forcing file.

The flux data files are not available on modelevaluation.org (unlike the meteorological forcing file). Instead, the flux data files for all calibration sites can be downloaded from https://github.com/callmip-org/Phase1/tree/main/Data/Phase1b.

Daily measurement errors are also provided to all groups based on FLUXNET processing (Pastorello et al., 2020). The uncertainty for each variable is designated by _uc. *These measurement errors need to be combined with modelling errors for the observation error covariance matrix in certain DA methods (e.g. variational)*. Each modeling group will decide on their own method for estimating model error.

Only measured flux data (i.e. non gap-filled, no energy balance closure corrected data) are provided. Days that contain more than 25% gap-filled half-hourly data points were excluded (set to NaNs). Days were also excluded if uncertainty estimates were not available. The daily missing % is provided in the flux data netcdf file for each variable.

The equivalent FLUXNET 2015 name for both the variables and their uncertainties can be found in the local attribute information within the netcdf file. For more information on these data please see Pastorello et al. (2020).

The same observation flux files in NetCDF format are provided to all groups for all selected sites. Individual modeling groups will then need to modify the files to work with their model/calibration system (for example, changing the variable names, adjusting or renaming the units and other attribute information).

NOTE: PFT type and % cover, soil layer depth and soil texture information is not provided in the global attributes of the flux observation file in Phase 1b. Instead, this information is provided in Tables 1 (calibration) and Table 2 (validation in Section 2.3 of the Protocol.

6. Required Output Variables and global attribute information

6.1 Output simulations

- Each group will conduct two simulations per site for each Calibration Scenario they perform (note: Scenario 1 is required – see Section 3.1) to ensure a consistent evaluation of model performance before and after calibration:

i. **prior simulation**, representing an out-of-the-box simulation using default model parameters in the model version being used at each site (i.e. no prior parameter tuning or updating of the model to fit the data). This will serve as a baseline to assess the impact of calibration.
ii. **posterior simulation**, using the optimized parameters and reflecting the improvements achieved through the calibration process.

- All groups must follow the spin-up and transient protocol prior to performing the parameter calibrations (see Section 8).

Optional:

- Groups using ensemble methods can submit prior and posterior ensemble simulations; however, this is not required for groups not using ensemble methods. The choice of how many ensembles to include, and the method for selecting those ensembles, is left to each model group.

- Groups may perform the calibrations with different model versions. In that case, we ask model groups to submit all required outputs for each model version and follow the file naming specifications listed in Section 7.

- All groups must output files according to output file specifications listed in Section 7.

6.2 Output variables

- All groups must output the following variables (per grid cell, not per PFT, age cohort or otherwise) with variable names according to the output file specification (Table 3) below:
    - NEE
    - LE
    - H
    - GPP
    - Total Ecosystem Respiration
    - Transpiration
    - Bare Soil Evaporation
    - Ground Heat Flux
    - Land Surface (Skin) Temperature
    - LAI
    - Total Aboveground Biomass
    - Total Soil Carbon
    - Total Column Soil Moisture
    - Top 10cm soil moisture (new in Phase 1b)
    - Latitude and longitude (see below)

A list of the variable CMIP short names, long names, units, and other notes are provided in Table 3. The naming format followed will be CMIP format from Phase 1b onwards with the exception of variables that are not standard variables in CMIP (see notes in Table 3 below). All variables are of the (time, lat, lon) dimension format as in the CMIP6 variable database here: https://clipc-services.ceda.ac.uk/dreq/mipVars.html. Please see the CMIP6 variable database for further information including standard names and descriptions, or ask Raj and Natasha.

Variable short (CMIP unless notes say otherwise)	**Variable long**	**Units (units attribute written exactly as shown below)**	**Notes**
nep	Net ecosystem productivity	kgC m-2 s-1	***Note: opposite sign from NEE that is used for calibration. nep* = positive flux is into the land**. Emissions from natural fires and human ignition fires as calculated by the fire module, but excluding any CO_2 flux from fire included in fLuc (CO_2 flux to the atmosphere from LUC). CMIP variable
hfls	Latent heat flux	W m-2	CMIP variable
hfss	Sensible heat flux	W m-2	CMIP variable
gpp	Gross primary production	kgC m-2 s-1	CMIP variable
reco	Total ecosystem respiration	kg m-2 s-1	reco = ra + rh Not CMIP variable (nor ALMA) though ra and rh are CMIP variables
tran	Transpiration	kg m-2 s-1	CMIP variable
evspsblsoi	Evaporation from soil	kg m-2 s-1	Includes sublimation. CMIP variable
hfg	Ground heat flux	W m-2	Not an official CMIP

			variable. Equivalent to Qg in ALMA format. Output only if your model outputs this variable.
ts	Land surface temperature	K	Surface (skin) temperature. CMIP variable
lai	Leaf area index	1 (or m2 m-2)	CMIP variable
cLiveBioAbove	Carbon mass in Above-ground Live Biomass	kg m-2	cLiveBioAbove = cLeaf + cStem Neither ALMA or CMIP (variable for iLAMB)
cSoil	Carbon mass in soil pool	kg m-2	CMIP variable
mrso	Total soil column moisture content	kg m-2	Mass of water in all phases (summed over all soil layers) per unit area. CMIP variable
mrsos	Moisture in upper portion of soil column	kg m-2	The mass of water in all phases in a thin surface soil layer (top 10cm). CMIP variable

Table 3: Variables to be output in all simulations for CalLMIP Phase 1b, including their short and long names, units and other notes.

- **Latitude and Longitude**: must be included as coordinates and as variables in each file. Dimensions of each variable should be (time, lat, lon)

  Naming format for lat/lon:

```
double lat(latitude) ;
        lat:units = "degrees_north" ;
        lat:long_name = "latitude" ;
double lon(longitude) ;
        lon:units = "degrees_east" ;
        lon:long_name = "longitude" ;
```

Note: lat/lon should be time independent, scalar values (i.e. single float, not arrays or strings)

No other formats are accepted in ME-org (i.e., X/Y, lndgrid, south_north/east_west, latitude/longitude will not be accepted)

ncdump/similar should produce something like:

```
dimensions:
	time = 6574 ;
	lat= 1;
	lon=1;
variables:
	double NEE(time, lat, lon) ;
…
```

- **Time**: All the flux files from ME-org contain daily data, including leap days. The output produced by each model should include leap days and should match the number of days present in the flux observation file provided in the git repository (see Section 5 (also see example output .nc file here: https://github.com/callmip-org/Phase1/tree/main/template_output_file).

  Models that do not include leap days are free to choose their preferred way of including leap days (e.g., avg of Feb 28 and March 1)

  Time format accepted: *"days since <first day of available FLUXNET data>"* e.g., "days since 1997-01-01 00:00:00".

  The daily increments should be 1 day: e.g., 0, 1, 2, 3,…, or 1, 2, 3, 4,... or 0.5 days: 0.5, 1.5, 2.5, 3.5,..., but not 1.45833333333333, 2.45833333333333, 3.45833333333333... or similar.

- **All groups must also provide posterior uncertainties** (or ensemble simulations) **for variables optimised** (NEE, LE, H) within the model output file. Use the suffix _post_unc for the variable name (e.g., nep_post_unc).

- Model groups do not need to provide posterior parameter values or their uncertainties. However, we ask that modeling groups retain this data in case we (as a community) decide that this information could be useful for further analysis or discussion.

6.3 Required global attributes

All groups must provide the following key-value pairs in the global attributes of all model output files

- Model: <model_name>.<model_version>
- CalLMIP_Phase: Phase1b
- Calibration_Scenario: <Scen1><Scen2><Scen3><Scen4><Scen5><Scen6>
- Calibration_stage: <Prior> or <Posterior>
- Cal_Val: <Calibration> or <Validation>

E.g., for the example template CLASSIC file provided here https://github.com/callmip-org/Phase1/tree/main/template_output_file and described in Section 7, the global attributes would be:

Model: CLASSIC.v2
CalLMIP_Phase: Phase1b
Calibration_Scenario: Scen1
Calibration_stage: Prior
Cal_Val: Calibration

Each group must also provide detailed metadata alongside their results. This information will be input directly into a draft paper documenting the 1st phase protocol, which will be sent to all participating groups when they sign up. Information required is listed in Section 12 and in the relevant methods and supplementary sections of the draft protocol paper.

7. Output file specifications

Model results must be output at daily frequency and stored in NetCDF files, using standard CMIP variable naming convention (see Table 3 in Section 6.2), and CF-compliant metadata, ensuring consistency in variable naming, units, and documentation. These files will then be uploaded to modelevaluation.org (see Section 11.5).

All output variables (optimized and additional – see Section 6) will be stored in the same output file as well as uncertainties on optimized variables.

A standardized naming system for output file names is required:
<*model_name*>.<*model_version*>_Phase1b_Scen<*calibration_scenario_number*>_<*Sit*

eName>_<Cal/Val>_<Prior/Posterior>.nc

For example, the example template output .nc file containing (see here: https://github.com/callmip-org/Phase1/tree/main/template_output_file) contains CLASSIC's Phase 1b prior simulation for the DK-Sor site for the required Calibration Scenario 1 and the filename is: CLASSIC.v2_Phase1b_Scen1_DK-Sor_Cal_Prior.nc

Note 1: "Phase1b" has been added to the filename for Phase 1b experiments to distinguish from Phase 1a (for DK-Sor).

Note 2: "CalLMIP Experiment" number in the filename has been replaced with "Calibration Scenario" number to avoid confusion with the CalLMIP Experiments (Phase 1a - test, Phase 1b Calibration and Phase 1b Validation) on ME-org. The same keyword is used in the global attributes (see Section 6.3).

Note 3: Cal/Val are a bit redundant as different sites are being used for calibration and validation, but we are keeping this information in the filename (and in the global attributes) for the sake of clarity.

Files containing global simulations (see Section 10) will have the following naming convention:

<*model_name*>.<*model_version*>_global_<*resolution*>_<*output_frequency*>_<*YYYY-YYYY*>_<*Prior/Posterior*>.nc

8. Spin-up and transient protocol for the parameter optimizations

Each modeling group must ensure that a full spin-up is performed and that models start from a stable, physically realistic state, preventing biases from arbitrary initial conditions. To keep the spin-up consistent across all models we are following the PLUMBER2 spin-up protocol for carbon (unless the group has a set functionality to prescribe initial conditions/states).

Models will cycle over the flux tower meteorological forcing files until equilibrium has been reached in the carbon stocks (and N and P included) (likely 100s to 1000s of years depending on model structure). We will spin up to equilibrium in carbon stocks representing the year 1850 using atmospheric CO_2 concentration and N deposition levels of 287.42 ppm and 0.79 kg N ha-1 yr-1, respectively. The transient simulation will cover the year 1851 to the first year of the forcing data. Climate forcing during this transient simulation will continue to cycle over the available flux tower site meteorological forcing files, but models should use historical yearly changes in atmospheric CO_2 concentration using the TRENDY CO2 file (see Section 4).

Note 1: If groups can and want to initialize their model runs by prescribing initial *conditions and states they can do that.*

Note 2: Forest age/disturbance that may be required for example by vegetation demography/cohort models is not provided. If needed, we can discuss further with relevant modeling groups.

9. Validation at site-scale

No temporal validation using the last year of data will be done in Phase 1b. Spatial validation will be conducted using the sites listed and described in Section 2.3.2. Meteorological forcing files are available for the spatial validation under the Phase 1b Validation experiment on ME-org. No flux data will be provided. Once model groups have completed the multi-site calibration, please run the simulations for the validation sites using both the prior (default) and calibrated parameters (for all calibration scenarios carried out in Section 3.1).

Model outputs and output file specifications must be the same for the validation runs as for the calibration (see Sections 6 and 7).

Note 1: Additional evaluation of variables not provided for optimization (as opposed to just an assessment of reduction in inter-model spread) will depend on data availability at the selected sites. If CalLMIP participants know of in situ data on model states (e.g., soil moisture, land surface temperature, aboveground biomass or soil C stocks) that are readily available, please let us know.

10. Post-calibration global simulations

Global scale runs (for both prior default parameters and posterior optimized parameters) will be performed after the Phase 1b full multi-site parameter calibration experiment is complete. These global scale simulations will allow us to evaluate model performance over the larger scale regions/continents that we typically look at in climate change impact studies.

Similarly to the site-level experiments, the input data for the global simulations, such as climate forcing, atmospheric CO_2, land use/cover change, N deposition etc will be the same across models and will follow the TRENDY protocol – i.e. we will use the TRENDY input data and the same spin-up and transient protocol). However, we will not perform all the simulations that are run for TRENDY. We will ask modelers to perform the equivalent of the TRENDY ‘S3’ experiment with all global change drivers (CLIM, CO_2, LUC, N DEP/FERT) and also a simulation that best matches the way the transient simulation was run for the site-level calibrations and validations – with CO_2 only (or CO_2+LUC and/or N DEP/FERT if modeling groups included these global change drivers in their site-level simulations).

In September 2026 we will send a separate document outlining the protocol for the global simulations. We will likely use TRENDY v15 forcing data that will be made available on Mike O'Sullivan's GitHub site (https://mdosullivan.github.io/GCB/) soon (June 2026).

Outputs will be standardized across models including the variables required (see Section 6), time period (likely at least the last 20 years), frequency of output (likely monthly and/or annual), and the spatial resolution (0.5 degrees, 1 degree or 2 degrees). These will be decided upon during a virtual meeting with CalLMIP participant model groups during the summer 2026.

The model outputs will follow the same variable and file format naming conventions as specified in Section 7 and will be uploaded to ME-org in a separate experiment "Phase 1 Global Sims" and evaluated using the iLAMB model benchmarking package which is integrated into ME-org.

Note 1: We note that these additional simulations require different forcing data to what will be used for the optimizations. We therefore will need to be careful not to discuss model "improvement" per se as biases/uncertainties due to parameters calibrated to different forcing may propagate in unknown ways to global scale or future simulations. What we can assess is the difference between prior and posterior and how that compares to benchmark data (for global runs) or how that changes future predictions. If participating model groups are interested, we may run an additional set of site level calibrations with the TRENDY forcing data to assess what impact the different forcing has on parameter calibration performance.

Note 2: We caution that global runs when model states have not been optimized is problematic and needs to be addressed in the optimization and/or considered when performing the additional runs/analyses.

11. Getting set-up on modelevaluation.org (ME-org)

11.1 Creating an account and accessing the CalLMIP workspace

You will need an account on modelevaluation.org (ME-org) to be able to access the CalLMIP workspace. If you haven't already, please create an account directly on the website.

- The CalLMIP workspace is public and therefore should be automatically shared with you once you sign up to ME-org.
- To find the CalLMIP workspace find the button in darker blue to the right of tabs at the top of the page titled 'Current Workspace'. Once you click the button you

will be taken to a new page where there are workspaces that have been shared with you.
- Scroll down to find CalLMIP.workspace to Click the CalLMIP workspace and you should see that darker blue tab at the top of the page change to say ‘Current Workspace: CalLMIP’. You should now be able view the data, sites, experiments and model output options available within that workspace.

11.2 Creating a Model Profile

All model outputs will be uploaded to ME-org. Every participating model group will need to create a new model profile.

- To create a new model profile, click on the ‘Model Profiles Tab’ and then select ‘Create New Model Profile’ from the drop down.
- Input Model name (and version if you are only submitting simulations from one model version)
- Select whether your model profile is visible publicly or only within the CalLMIP workspace.
- Provide a url(s) for the model.
- Provide relevant references for both the model and calibration system. However, please note that we will collect most of the information about the model and calibration within the Phase 1 protocol draft paper. See Section 12 for details that will be required.
- Provide any additional comments that you think may be helpful, but see note in previous bullet point about where we collect most of the information relating to the model and calibration system.
- Click save and now this model profile can be repeatedly accessed within the ‘Model Outputs’ tab on the top of the page. You can upload as many model outputs (including prior and posterior, calibrated and validated, required and optional calibration scenarios, different versions) under this created model profile, as needed.
- Note: all participating model profiles are available to view under ‘Model Profiles’ → ‘In Current Workspace’.

11.3 Finding CalLMIP experiment information on ME-org

Information on all experiments being performed within CalLMIP Phase 1 (Phase 1a - test, Phase 1b Calibration and Phase 1b Validation) can be found within the CalLMIP workspace on ME-org (see Section 11.1 to create an account on ME-org and find the CalLMIP workspace).

- To access the information on the Phase 1a/b experiment click on the 'Experiments' tab on the top of the page and go to 'In Current Workspace' on the drop down menu.
- There are two new experiments for Phase 1b: Phase 1b Calibration (which includes all the site pages/met forcing files to use in the multi-site calibrations) and Phase 1b Validation (which includes all the additional sites to use for the "spatial" validation).
- Click on each of the CalLMIP Phase 1b experiments from the list to see general information about the experiment (short and long description), the Data Sets (sites) attached to each, and the model outputs already uploaded to the experiment.
- Note that the empirical models – Gab Abramowitz's machine learning 'out-of'sample' models for each site – that are used for benchmarking (see Section 11.5) in addition to the observations are also listed under the Model Outputs for each experiment (3 models of varying complexity labelled "Emp 1lin", "Emp 3km27" and "Emp LTSM"). See the Abramowitz et al. (2024) for further details.
- Additional information – such as how to partition Precip into Rainf and Snowf – is also provided on these experiment pages.

11.4 Finding site and data information on ME-org

Information on all sites and forcing files used within the CalLMIP project can be found within the CalLMIP workspace on ME-org (see Section 11.1 to create an account on ME-org and find the CalLMIP workspace and experiments (Section 11.3)).

- To access site information and meteorological forcing datasets available for each site click on the 'Experiments' tab at the top of the page and then click on either 'Phase 1b Calibration' or 'Phase 1b Validation' to see a list of the Data Sets (or sites – half way down those pages). Click on each site name to take you to a page with information and plots for each site as well as the met forcing files to download.
- (Note you can also access all the sites associated with CalLMIP – for the calibration and validation runs – by clicking on the 'Data Sets' tab on the top of the page and go to 'In Current Workspace' on the drop down menu).
- There are 21 sites in the Phase 1b Calibration experiment and 10 sites in the Phase 1b Validation (see Section 2.3 for further details on these sites).

- *Note 1: Flux observation files are NOT available on ME-org. The flux data will be shared via the CalLMIP GitHub site. See Section 5 for further details.*

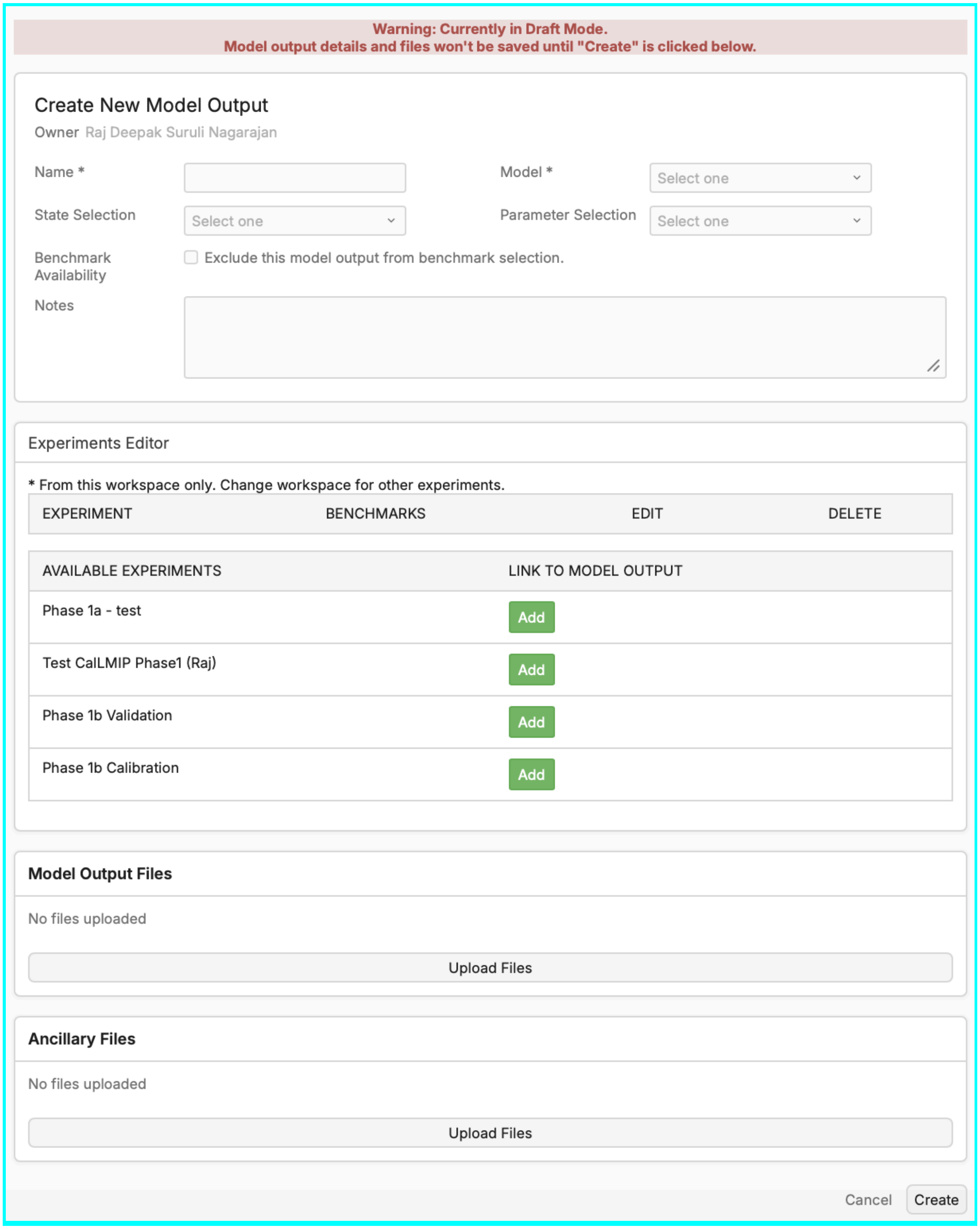
11.5 Uploading Model Outputs on ME-org

Once you have created a model profile (see Section 11.2), and have completed the Calibration Scenario 1 required for Phase 1b, you can then upload your model outputs.

- For both calibration and validation you will upload all your prior runs for all sites for each calibration scenario you perform as one model output (i.e., prior runs for all calibration sites for Scenario 1 as one “Model Output” upload), and your posterior runs for all sites (calibration or validation) for each calibration scenario you perform as another model output, etc.
- As you upload each set of model outputs you will link them to a CalLMIP Experiment (Phase 1b Calibration or Phase 1b Validation).

<u>To upload your model outputs for the prior and posterior for each Calibration Scenario</u>:

1. Click on the ‘Model Outputs’ tab at the top and select the option to ‘Upload Model Output’. You should now see this page:

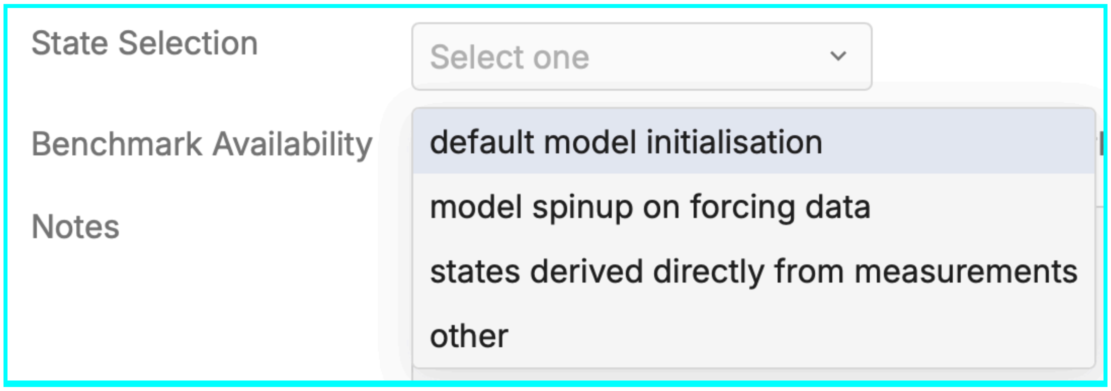


2. Under “Name”, provide the following: *<model_name>.<model_version>*_Phase1b_Scen<*calibration_scenario_number*>_<*Cal/Val*>_<*Prior/Posterior*>. I.e., everything that’s in the filenames except the

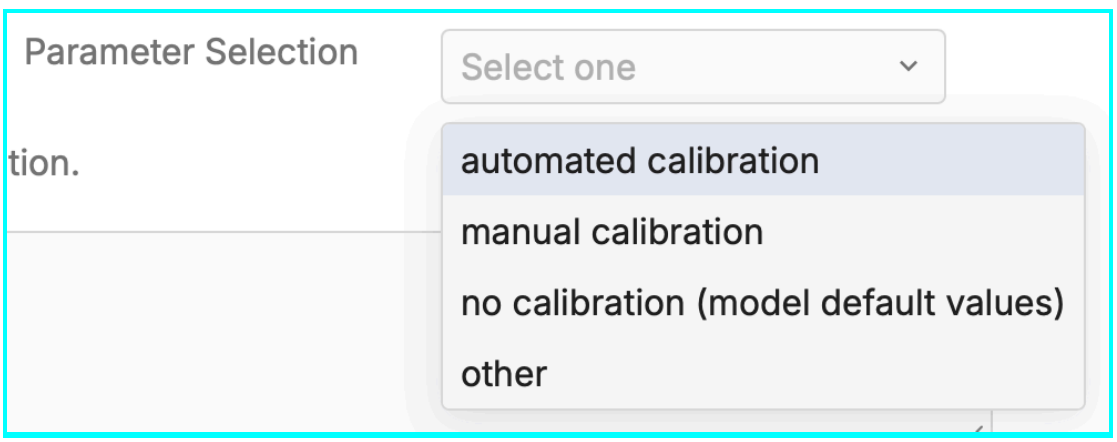
site name and the “.nc”. E.g., in the case of the example template .nc file provided here (https://github.com/callmip-org/Phase1/tree/main/template_output_file) the “Name” would be CLASSIC.v2_Phase1b_Scen1_Cal_Prior

3. Within the ‘Model*’ dropdown you should be able to see the model profile you have previously created from the ‘Create New Model Profiles’ (see Section 11.2).

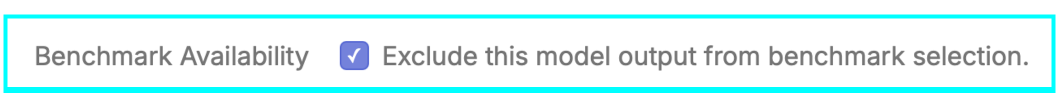
4. Under ‘State selection’ select the relevant description. If your method is not listed please select ‘other’ and ensure this is explicitly detailed within the comments section at the bottom of the page.



5. Under ‘Parameter Selection’ select the relevant description (no calibration for prior; automated calibration for posterior).



6. Check the “Benchmark Availability” checkbox. Only the Empirical models from Gab’s machine learning models (shown as “Emp_*” under Model Outputs) will be used for benchmarking, *note* other land models.

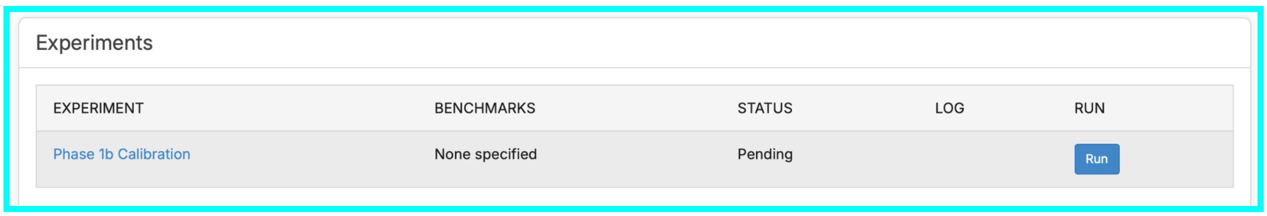


7. From the list of “Available Experiments”, “Add” the appropriate Experiment(s) to “Link to Model Output” depending on whether the upload is for calibrationor validation, as follows:
    - Phase 1b Calibration; or
    - Phase 1b Validation

8. Upload .nc files for all the sites for each Calibration Stage (prior/posterior) for the required Calibration Scenario 1. For each stage (prior/posterior), you should have 21 files for calibrations, and 10 files for validations. Repeat for any optional additional calibration scenarios you perform.

9. **Press ‘Create’ at the bottom of the page to save this model output upload.**

If you click on/re-open the model output just created (by clicking on the Model Outputs tab at the top of the page), you will see under “Benchmarks” that it says “None specified” and that Status of the experiment is “Pending”.

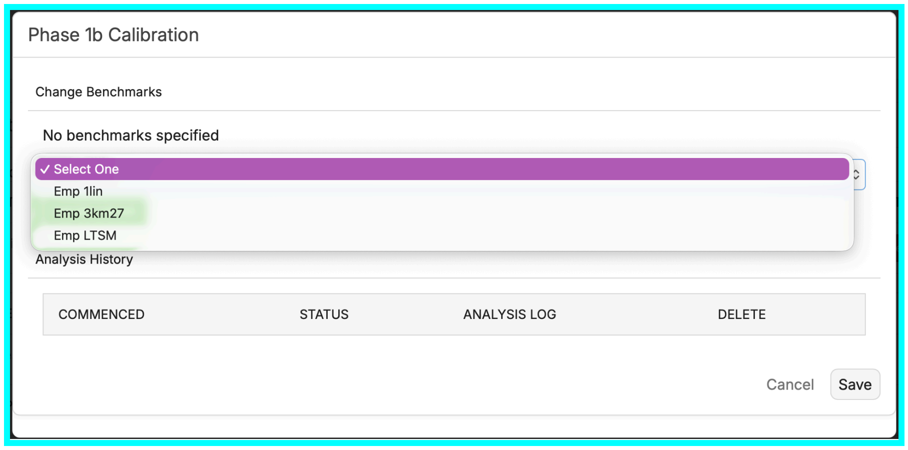


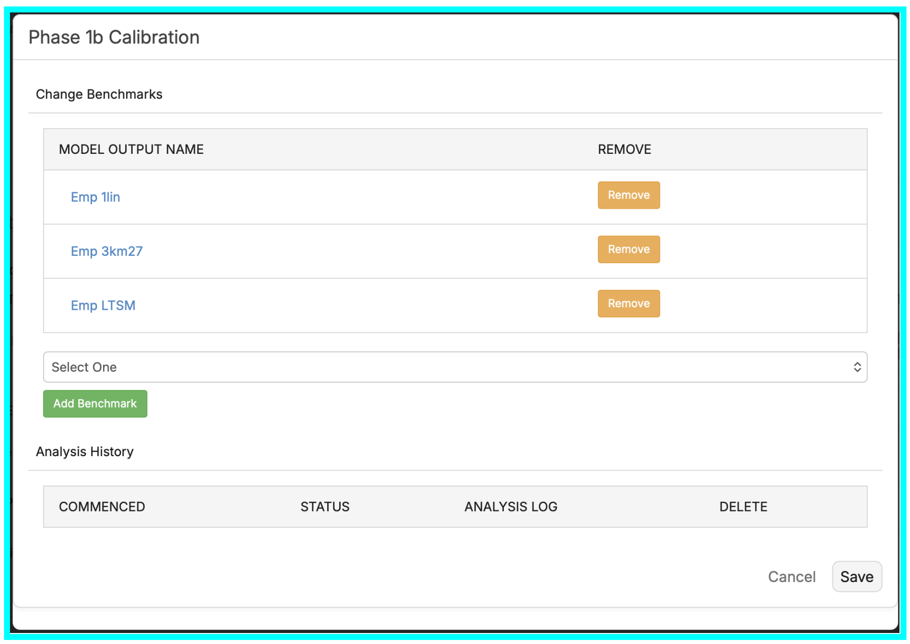
Please run the benchmarks (comparison to Gab’s machine learning models trained across sites) by completing the following steps:

10. Click on “Edit” on the bottom right corner of the model output page

11. Click on “Edit” within the experiment

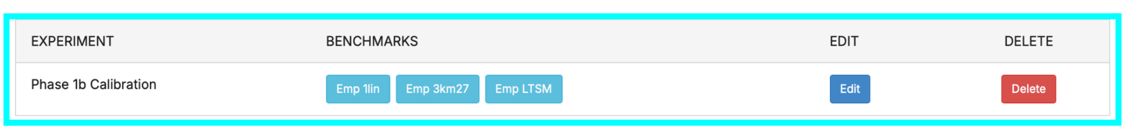
12. In the pop-up window, select the empirical models (Emp 1lin, Emp 3km27, Emp LTSM – see image below) to add them as benchmark models, one at a time, and “Add benchmark”





13. Click on "Save" at the bottom of the page.

The empirical benchmark models have been added to compare to your model outputs.



14. Click on "Save Changes".

15. Press "RUN" to run the comparison between your model outputs and the observations as well as the empirical benchmark models.

16. Check if you get an error message and/or if the plots generated through "View plots" have greyed out bars (which means your model outputs are not loading

properly). The error will most probably be related to something in the nc files (as below: time, lat/lon, units) so please check through Sections 6 and 7 again.

17. If errors persist even after the output formats from Sections 6 and 7 are followed, please contact Raj (rsurulin@uwo.ca) (Note: if you are working with monthly data, you will have an error message pop up. No need to contact Raj in this case. These outputs will be processed and analysed offline).

18. Repeat for the prior and posterior simulations for the calibration and validation and for each Calibration Scenario, if your group decides to perform the optional Calibration Scenarios 2-6.

12. Model information table

The following information will be required from each modeling group. This information will be input directly into a CalLMIP 1st Phase protocol draft paper that we aim to publish in GMD (see Proposed Timeline). This draft paper will be shared with all participating model groups once they have registered on the CalLMIP website. The information required includes (but please see the draft paper for the most up-to-date list):

i. Brief description of the model including the model version(s) plus all relevant references.
ii. Further to point (i), are you including N cycling, fire, dynamic vegetation…? (if so, please provide references
iii. If you do include N cycling, which N deposition and N fertilisation (for crop sites) are you using?
iv. What parameter estimation method are you using? Detailed steps and equations should be documented in your models' section below. Any other details on configuration/set-up should also be included (e.g., no. of ensembles etc).
v. Description of parameter selection method (including SA if SA was performed) and any additional screening steps that were applied. Include how many parameters were selected out of a total X number of possible model parameters to be calibrated.
vi. Table of parameters included in the calibration, including their short (model) and long name, description, process associated with the parameter, prior/default value, prior uncertainties, ranked sensitivity (if an SA was performed).
vii. Method for the spin-up beyond what is outlined above?
viii. Method for accounting for potentially biased initial conditions (soil C stocks etc) - are they included in the parameter calibration or prescribed for example?
ix. Method for calibrating with multiple datastreams (scenarioss iv to vi) what is your method/approach for multi data stream assimilation(e.g., step-wise vs simultaneous)? I.e. How are these datasets combined in the cost/objective function?
x. What is your method for generating an operational set of parameters for running the model globally? (e.g. multi-site optimisation per PFT or parameter ensembles?)
xi. Do you calibrate all PFTs for a given site or only the most dominant PFT (where applicable)?
xii. What other sources of uncertainty are being included/propagated in the calibration (e.g., initial condition, driver, boundary, demographic stochasticity, random process error).

xiii. Which optional scenarios are you participating in (e.g., NEE only, LE only etc; see protocol Section 3.1).
xiv. Did you prescribe LAI from the satellite time series provided in the forcing file, or did you use a prognostic LAI?

13. Open Data Policy

All outputs from CalLMIP parameter calibration experiments are hosted on ME-org. These model outputs will be freely available for public use after 1 year of publication under the Creative Commons Attribution 4.0 International (CC-BY-4) license by registering on ME-org. Use of model outputs must include an acknowledgement of the model group that generated the model outputs and citation of the relevant model specific and CalLMIP papers. The full acknowledgement and citation policy (including model specific acknowledgements/references) will be provided on the CalLMIP website, in the Phase 1 protocol paper, and in the public repository.

14. CalLMIP co-authorship policy

We recognise that global land surface model parameter calibration systems often require a considerable number of people – often early career researchers – for their development and indeed for carrying out the simulations/calibrations required by this voluntary CalLMIP initiative. Therefore, our primary goal is to ensure all those who contributed significantly to enabling CalLMIP to happen will be rewarded by co-authorship on CalLMIP papers.

Any individual who contributes to performing parameter calibration experiments, and/or who contributes in an intellectual capacity to the design of the experiments and analyses, and/or who advises individuals carrying out parameter calibration experiments, and/or who documents the experiment information, will be included as a co-author on papers resulting from the CalLMIP 1st Phase experiments. Co-authors will also be expected to provide feedback on the structure and framing of the paper as well as on draft(s) of the paper.

For each of the CalLMIP papers we are working on a more detailed ‘CRediT’ contributor matrix (‘Contributor, Roles, Taxonomy as columns and people as rows) with co-authorship requiring a certain number of those items in the coming weeks (November 24th 2025), which we will share with participating groups as they sign up. However, we note that we want to offer participating model groups as much autonomy and flexibility as possible when deciding who should be a co-author on the paper, while encouraging transparent, merit-based decisions that recognize all career stages.

This policy applies to the Phase 1 protocol paper, the Phase 1 site calibration synthesis paper and the Phase 1 global simulations paper. Additional papers may have different authorship terms determined by their scope.

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